

System Flows & Process Guide

Ordering · Delivery · Customer · Seller · Rider · Super Admin

Prepared for the client

A visual, plain-language walkthrough of every business flow in the platform, with annotated flowcharts and step-by-step descriptions for each portal.

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Customer

Seller

Warehouse

Rider

Super Admin

Contents

1. Platform Overview	3
The six portals	3
Key business rules baked into the flows	3
2. End-to-End Order Lifecycle	5
How an order moves	5
3. Ordering Flow	7
Step by step	7
4. Delivery & Fulfilment Flow	9
Warehouse operations	9
Rider last mile	9
Exceptions	9
5. Customer Portal	11
6. Seller Portal	13
From sign-up to selling	13
7. Rider Portal	15
A rider's day	15
8. Super Admin Portal	17
What the admin governs	17
Appendix · Status Reference	19

1. Platform Overview

Somba & Tekka is a Flipkart-style online marketplace where many independent sellers list products and the platform coordinates payment, warehousing and last-mile delivery. The product is delivered as **six role-based portals** that share one codebase, plus two native mobile apps. Everyone signs in through a single door and is routed to the portal that matches their role.

The six portals

- **Customer Shop (/shop)** — browse, buy, track orders, returns, wallet. Also a Flutter mobile app.
- **Seller Portal (/seller)** — list products, fulfil orders, manage finances and payouts.
- **Warehouse Portal (/warehouse)** — receive, sort, batch and dispatch parcels; handle returns and exceptions.
- **Rider Portal (/rider)** — accept delivery tasks, navigate, capture proof of delivery. Also a Flutter app.
- **Super Admin (/admin)** — govern the whole marketplace: approvals, money, catalog, settings, oversight.
- **Landing & Auth (/ , /login)** — marketing entry and the shared sign-in for all 11 personas.

Key business rules baked into the flows

- **Dual market, one codebase** — a **France** profile (demo, USD) and a **DRC** profile (Kinshasa / Lubumbashi, USD with a manual USD→CDF rate). Every screen is bilingual **English / French**.
- **Seller-owned inventory, hybrid fulfilment** — goods can ship from the seller or from a platform warehouse. **Cross-city** and **open-box** delivery are supported.
- **Commission** is category-tiered (**8-15%**) with seller-tier discounts; seller earnings clear after delivery + warehouse reconciliation + a **48-hour** hold.
- **Payments are prepaid** — Stripe card, Airtel Money (mobile money) and the in-app **Somba Wallet**. *Cash on Delivery is not offered* — every order is paid before dispatch. Refunds go back to the original method or to the wallet.
- **Prototype scope** — the system runs on realistic **mock data**; there is no live backend, real payment capture or push notifications yet. The flows below describe the intended production behaviour.

Technology. All six portals are one Next.js web application; a NestJS service provides a mock API; the customer and rider mobile apps are built in Flutter; and a shared library holds the screen registry, market profiles and the status definitions that every portal agrees on.

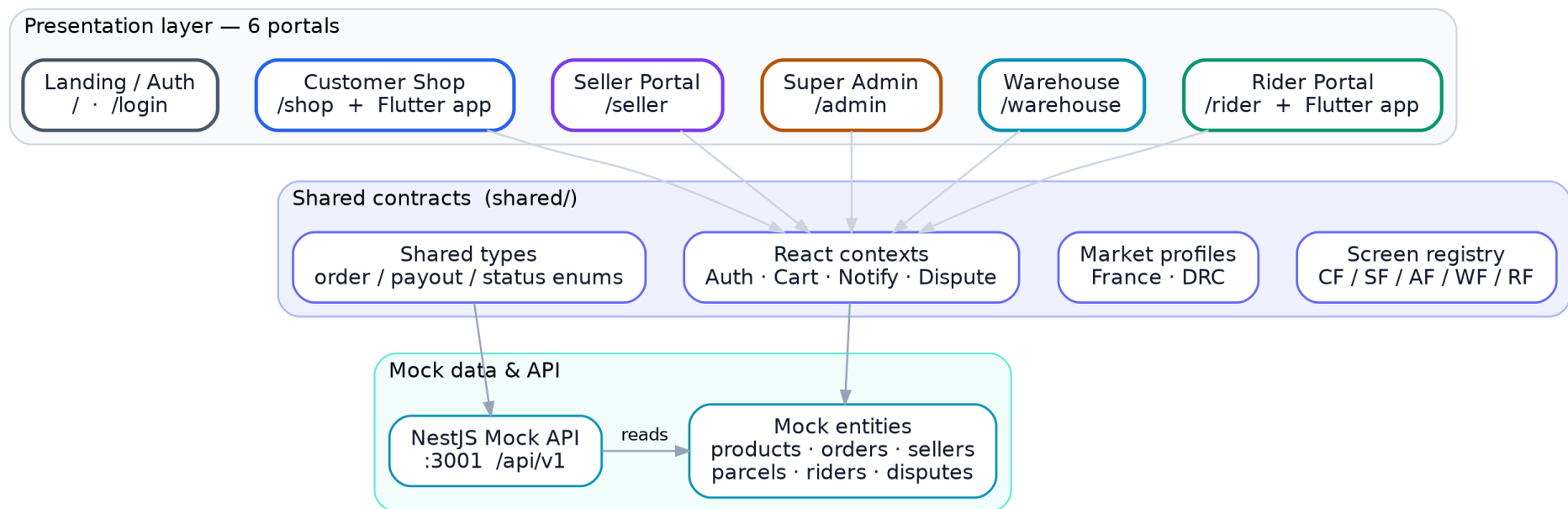


Figure 1 — System architecture — six portals share React contexts and a mock API/data layer.

2. End-to-End Order Lifecycle

This is the single most important journey: how one order travels across **all** portals, from the customer tapping “buy” to the seller being paid. Each colour in the flowchart is a different actor (customer, seller, warehouse, rider, admin), so the hand-offs between teams are easy to see.

How an order moves

- **Customer** places and pays for an order. Status goes **pending** → **confirmed**.
- **Seller** receives the order, accepts it (**processing**) and marks the package ready for pickup.
- **Warehouse** takes the parcel inbound, receives and scans it, sorts it to a zone (**ready**), then builds a delivery batch.
- **Warehouse** assigns a rider and dispatches the batch. Order goes **shipped** → **out for delivery**.
- **Rider** picks up, navigates to the customer and attempts delivery — the key decision point.
- On success the rider captures **proof of delivery** (OTP and photo / signature); order becomes **delivered**. On failure the reason is logged and the parcel returns for re-dispatch.
- **Seller** earnings clear (48h) and a payout is requested; **Super Admin** approves it and it is marked **paid** — the order is fully settled.
- At any point after delivery the customer may open a **return, exchange or dispute**, which the seller and admin resolve, with the admin authorising any **refund**.

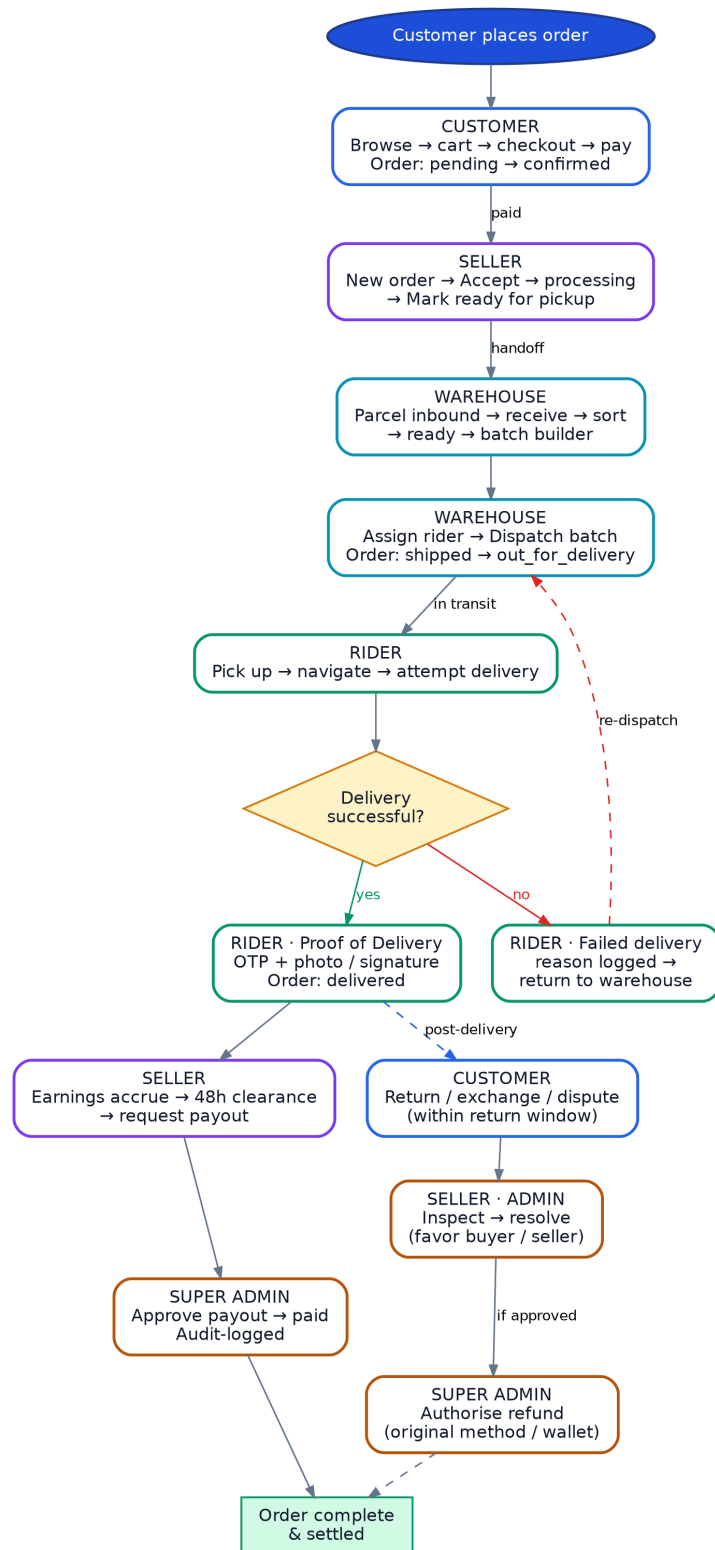


Figure 2 — End-to-end order lifecycle across all portals (colour = responsible actor).

3. Ordering Flow

The ordering flow is the customer's purchase funnel — everything from opening a product to a confirmed order. The checkout is a guided **four-step** process and the system applies market rules (such as zone delivery fees) as the customer proceeds. Every order is **paid up front**.

Step by step

- **Discover & add to cart** — from a product page the customer can pick a variant, check delivery to their pincode/zone, read Q&A and reviews, then add to the cart. The cart groups items by seller and supports quantities and “save for later”.
- **Promo code** (optional) — codes such as **SOMBA10** (10% off over \$50) or **SAVE20** (\$20 off over \$100) apply when the minimum-order rule is met.
- **Sign in or guest** — the customer can log in or check out as a guest with just an email.
- **Step 1 · Address** — choosing the delivery address sets the **zone delivery fee** automatically.
- **Step 2 · Options** — cross-city delivery, open-box delivery and an order note.
- **Step 3 · Payment (prepaid)** — the order is paid before dispatch by card, the **Somba Wallet** or **Airtel Money** (mobile money). There is no cash-on-delivery option, so no order leaves the warehouse unpaid.
- **Payment outcome** — if a payment fails the reservation is held for **15 minutes** and the customer can retry. On success the order is confirmed.
- **Step 4 · Confirmation** — the order is created with status **processing** and live tracking becomes available.

Payment methods at a glance (all prepaid)

Method	How it works	Availability
Stripe card	Card payment (mock Stripe element)	All orders
Somba Wallet	Store credit; top-up via Airtel Money	Up to balance
Airtel Money	Mobile money, phone-number entry	All orders

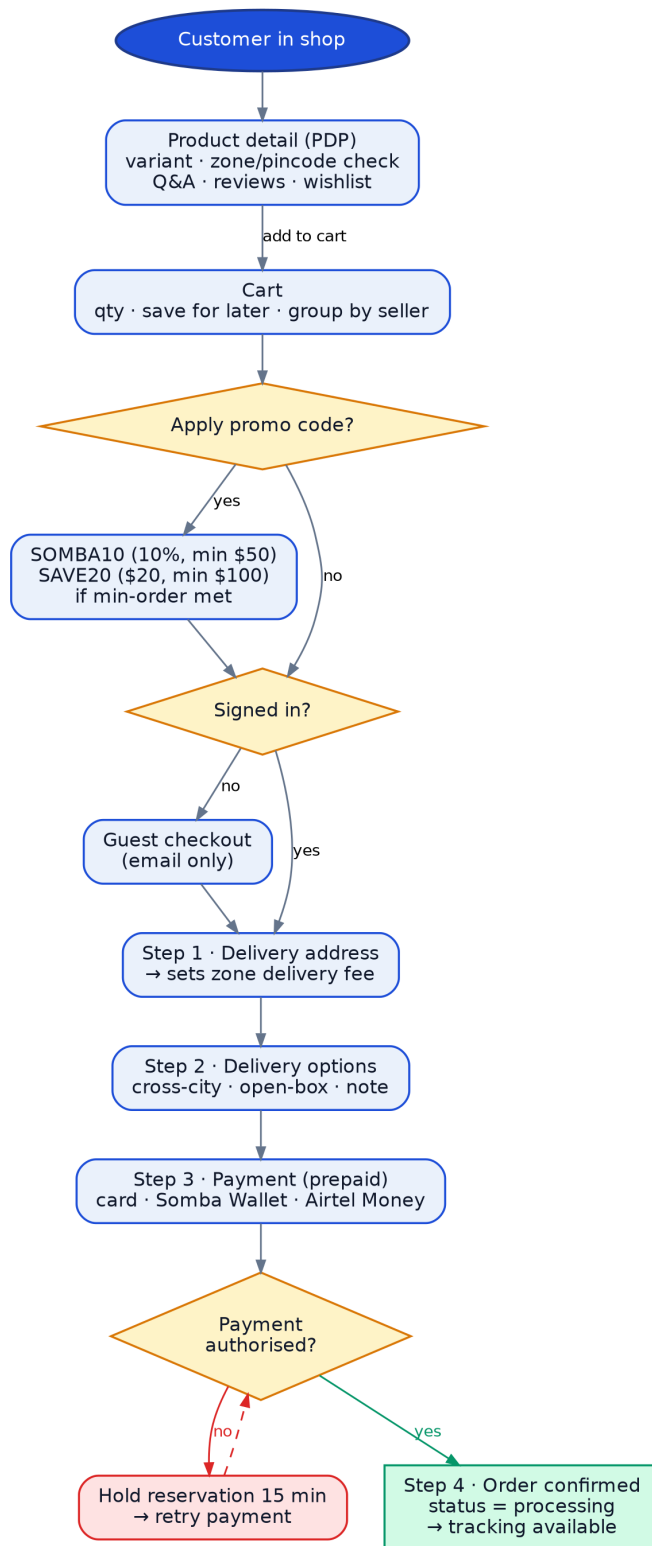


Figure 3 — Ordering flow — browse to confirmed order, with promo, guest and prepaid payment-retry branches.

4. Delivery & Fulfilment Flow

Once an order is paid, the operations side takes over. This flow has two halves: the **warehouse** prepares and dispatches parcels, and the **rider** completes the last mile. The same warehouse screens are also visible to the Super Admin under *Fulfilment*, and the platform supports **multiple hubs**.

Warehouse operations

- **Inbound** — parcels collected from sellers arrive at the hub.
- **Receiving** — staff scan the barcode and inspect. If inspection fails, an **exception/incident** is raised (damaged, missing item, count mismatch).
- **Sorting** — the parcel is assigned to a delivery zone and becomes **ready**.
- **Batch builder** — ready parcels are grouped into a delivery batch and the route is optimised.
- **Assign rider** — the system suggests the **nearest available rider** (or staff pick manually).
- **Dispatch** — the batch is handed to the rider; the order moves to **out for delivery**.

Rider last mile

- **Pick up & navigate** — the rider collects the batch (status **in transit**) and is guided stop-by-stop, able to call the customer.
- **Proof of delivery** — on arrival the rider verifies the OTP, captures a photo/signature and performs the open-box check where required (orders are already paid, so no cash changes hands).
- **Outcome** — success marks the order **delivered**; a failed attempt logs a reason and the parcel is returned for re-dispatch.

Exceptions

- **Incidents** — damage, missing items or count mismatches raise an exception that is investigated and resolved before the parcel continues.
- **Aged parcels** — anything stuck beyond ~72 hours is flagged and escalated.

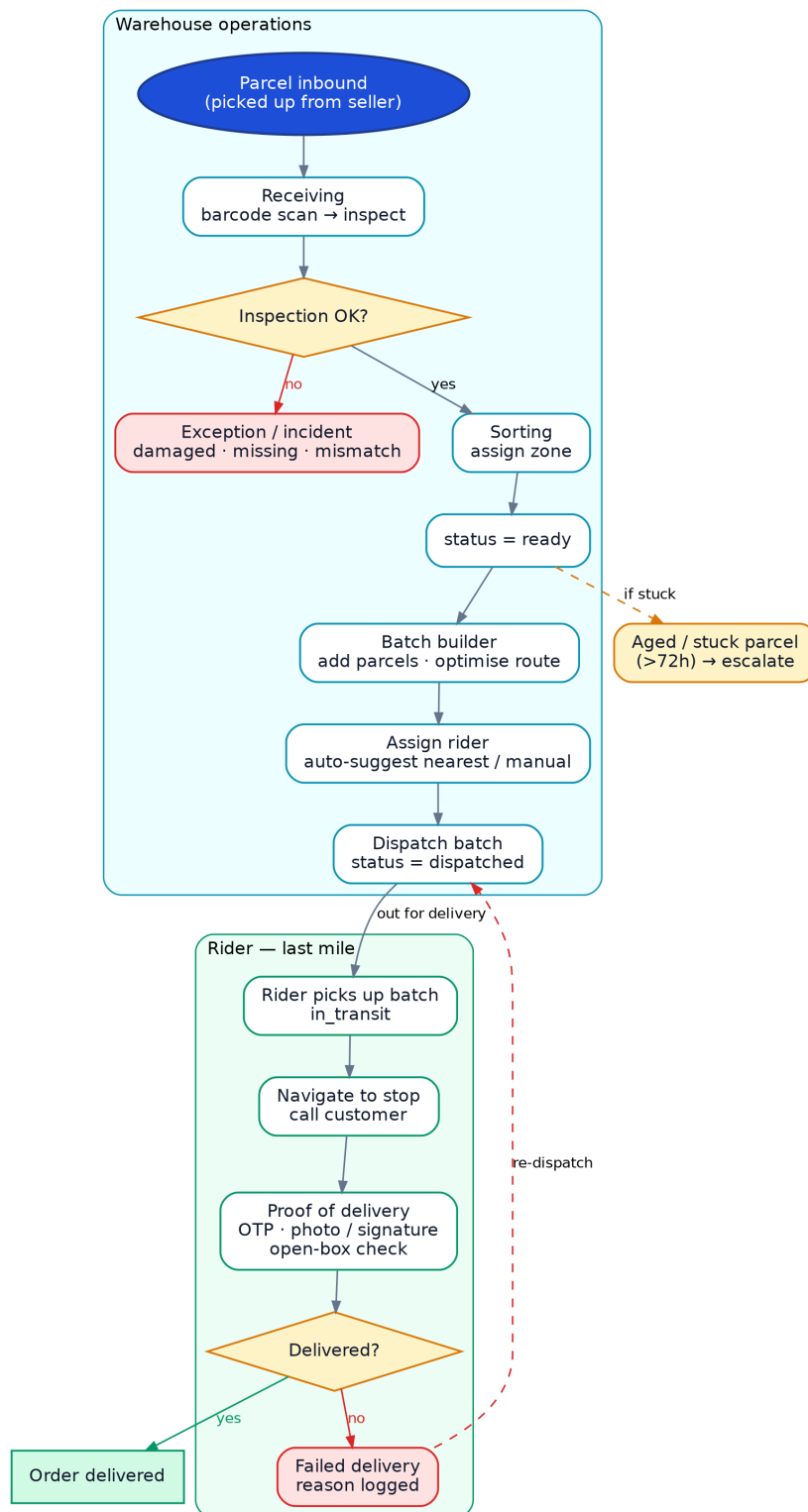


Figure 4 — Delivery & fulfilment — warehouse intake to dispatch, then the rider's last mile to a delivered order.

5. Customer Portal

The customer portal (web at **/shop** plus a Flutter app) is organised into five areas that map to a shopper's journey: **account**, **discover**, **buy**, **after-purchase** and the **account hub**. The ordering funnel in section 3 lives inside the "buy" area.

- **Account & auth** — register → OTP → verify email, plus login and forgot/reset password.
- **Discover** — a personalised home feed (buy-again, recently viewed), search with autocomplete, category browsing and rich filters, product and store pages with Q&A, reviews and store-follow, wishlist and a flash-deals hub.
- **Buy** — cart with promo codes, the four-step checkout, and order confirmation.
- **After purchase** — order history, **live per-parcel tracking** with map, one-tap reorder, and return / exchange / dispute requests with a support ticket channel.
- **Account hub** — the **Somba Wallet** (top-up and transaction history), **Refer & Earn** (\$10 / \$10), address book, notification centre, and help including account deletion.

Returns move through **requested** → **approved** → **in transit** → **received** → **refunded** (or rejected); disputes move through **open** → **seller responded** → **resolved** → **closed**. Both keep the customer informed at every status change.

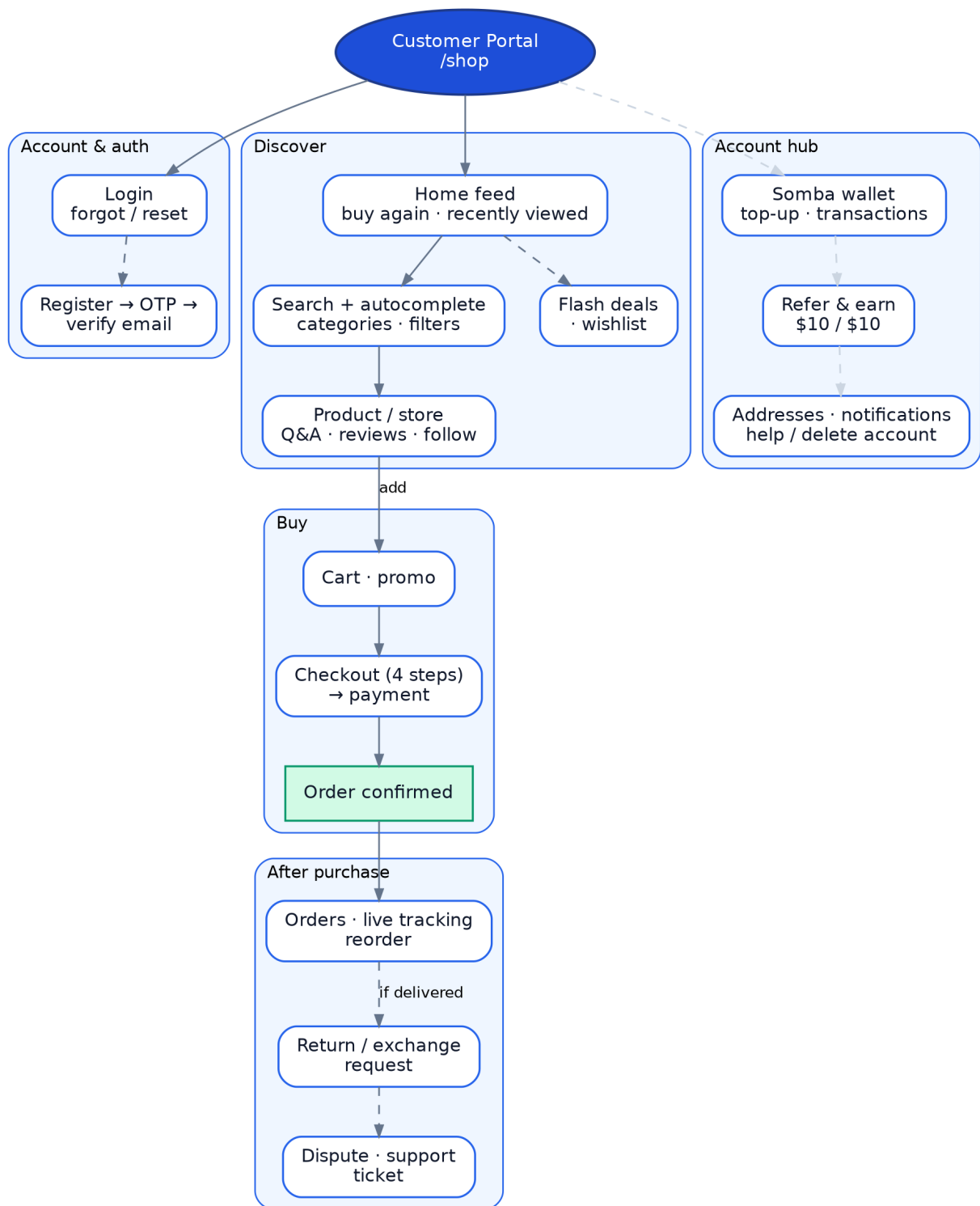


Figure 5 — Customer portal — the shopper journey from account and discovery through to after-purchase care.

6. Seller Portal

The seller portal is where a merchant runs their storefront: getting approved, listing products, fulfilling orders and getting paid. Listings and promotions pass through admin moderation before going live, which keeps catalog quality high.

From sign-up to selling

- **Onboarding** — from the sell-online landing the merchant chooses a plan and subscribes, registers their business (lightweight, **no KYC** in the prototype), and lands in a **pending approval** state. The admin approves, or the seller resubmits corrections.
- **Dashboard** — once approved, KPIs, a store-health badge (Gold / Silver / Bronze / Somba Assured) and a finance snapshot.
- **Catalog** — a guided **7-step product wizard** (basics → media → variants → inventory → pricing → shipping → review). On submit the product enters moderation (**draft** → **pending** → **approved/live**). Bulk CSV import/export and a flag-unavailable action are available.
- **Fulfilment** — new orders are accepted (**processing**) and marked ready; the package is handed to the warehouse and tracked. Returns, replacements and disputes are handled here too.
- **Money & growth** — transactions show commission (8-15%); earnings clear after delivery, reconciliation and a 48-hour hold, after which the seller **requests a payout** for admin approval. Storefront design, promotions, analytics and review replies round out the portal.

Seller payout — money states

Earning status	Meaning
awaiting_delivery	Order not yet delivered
pending_clearance	Delivered; inside the 48h clearance window
ready_for_payout	Cleared and available to withdraw
paid_out	Included in an approved, paid payout

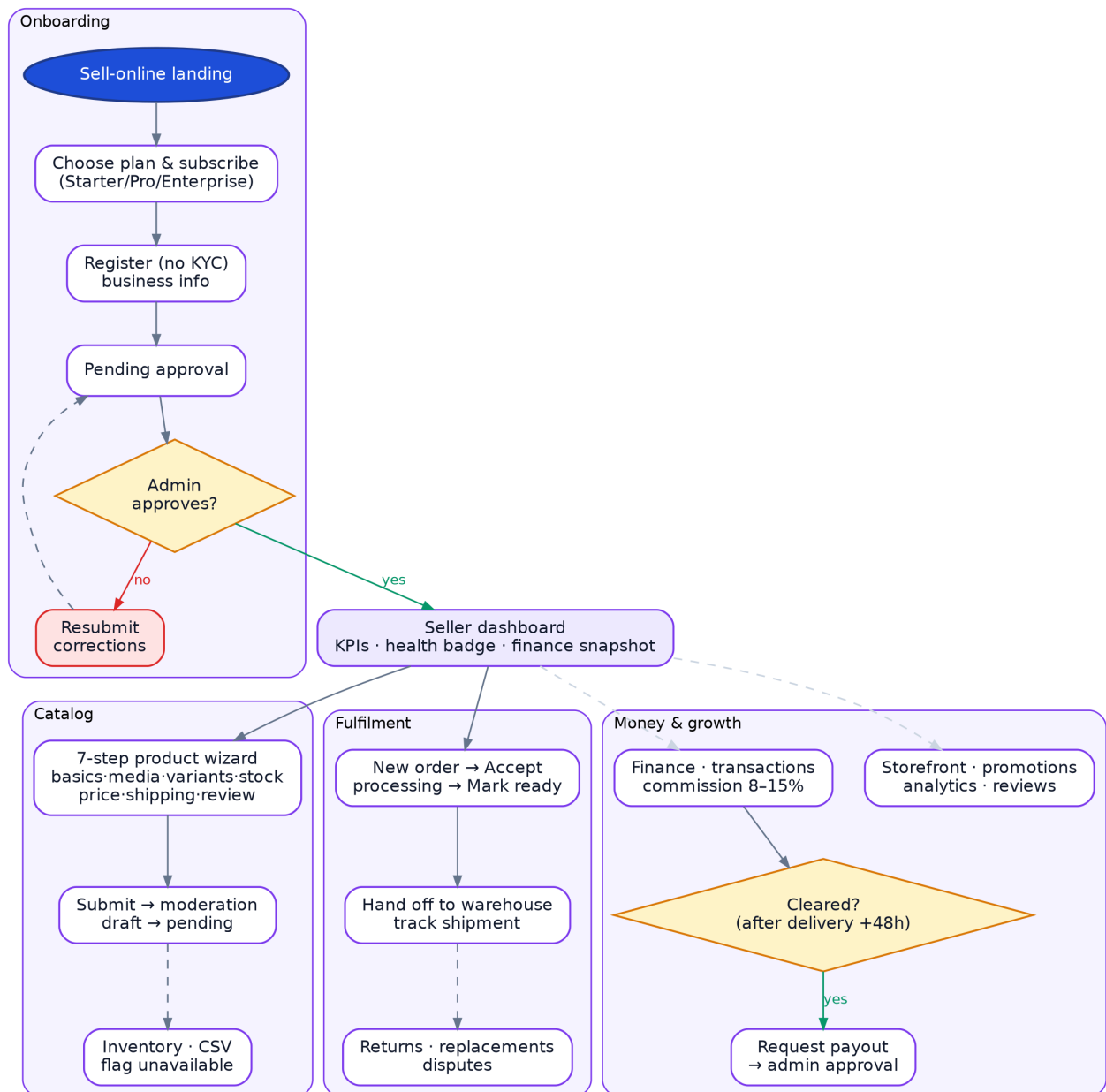


Figure 6 — Seller portal — onboarding, catalog and moderation, order fulfilment, and the payout pipeline.

7. Rider Portal

The rider portal (web at **/rider** plus a Flutter app) is the delivery agent's daily tool. It is built around a clear task lifecycle and a structured **proof-of-delivery** step that protects both the customer and the platform. Because every order is prepaid, the rider never handles cash.

A rider's day

- **Sign in & go on-duty** — on first login the rider sets a password, then toggles availability to start receiving work.
- **Receive & open tasks** — a *task-assigned* notification leads to the task list and a **batch overview** showing ordered stops; each task detail shows the order, customer and notes.
- **Navigate & pick up** — map navigation and one-tap call; the task moves **assigned → picked up → in transit**.
- **Proof of delivery** — enter the OTP and recipient name, capture a photo/signature and complete the open-box check where required. The order becomes **delivered**.
- **Failed delivery** — if it cannot be completed, the rider logs a reason (customer unavailable, wrong address, refused, other) and the parcel returns to the warehouse.
- **End of shift** — an earnings view shows per-delivery pay plus incentives, with shift history.

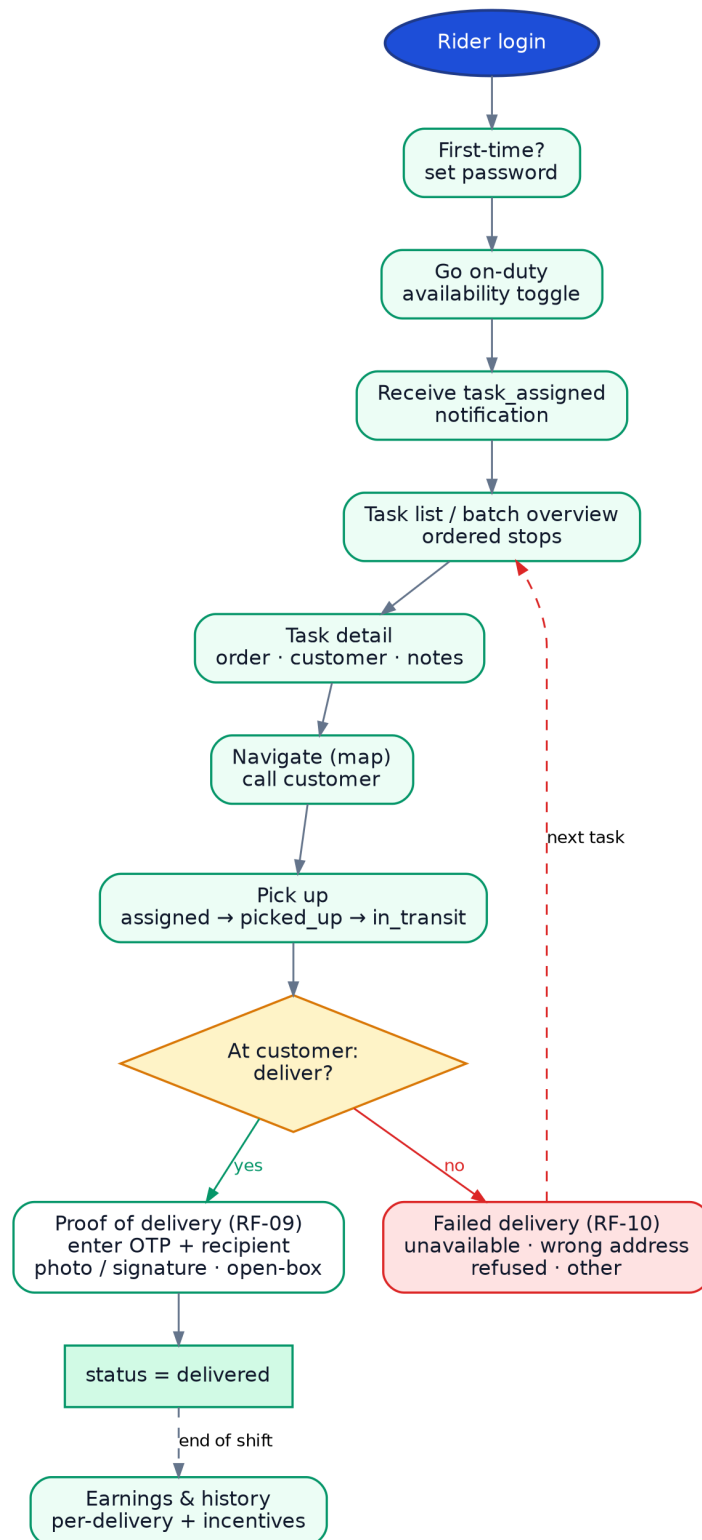


Figure 7 — Rider portal — from going on-duty through navigation and proof of delivery to a delivered order.

8. Super Admin Portal

The Super Admin portal governs the entire marketplace. Access is divided into **six sub-roles** (role-based access control), and **every action is recorded in an audit log** with before/after values, so the platform always has an accountable trail.

What the admin governs

- **Trust & approvals** — approve or reject sellers; moderate products (**approve / reject / request changes**) and reviews; and triage fraud & risk alerts (velocity, address blocks, OTP failures) through **open → reviewed → blocked**.
- **Money** — resolve disputes in favour of buyer or seller; authorise refunds (to original method or wallet); and approve seller payouts (**requested → approved → paid**, \$10 minimum, 48-hour clearance).
- **Catalog, marketing & platform** — categories and commission rules, flash sales and seller-promotion approvals, CMS homepage blocks and broadcasts; plus platform **settings** (dual-market, FX rate, delivery fees), delivery **zones**, and **warehouse** creation with auto-generated portal credentials.
- **Oversight** — the audit log, an analytics suite (GMV, orders, sellers, SLAs, marketplace health), and read-through into orders, customers and platform finance.

The six admin sub-roles

Sub-role	Primary responsibilities
Operations	Orders, warehouse, logistics
Finance	Payouts, refunds, reports
Support	Tickets, customers, returns
Marketing	Campaigns, CMS, coupons, banners
Moderation	Products, reviews, sellers
Warehouse	Inventory, dispatch, hubs

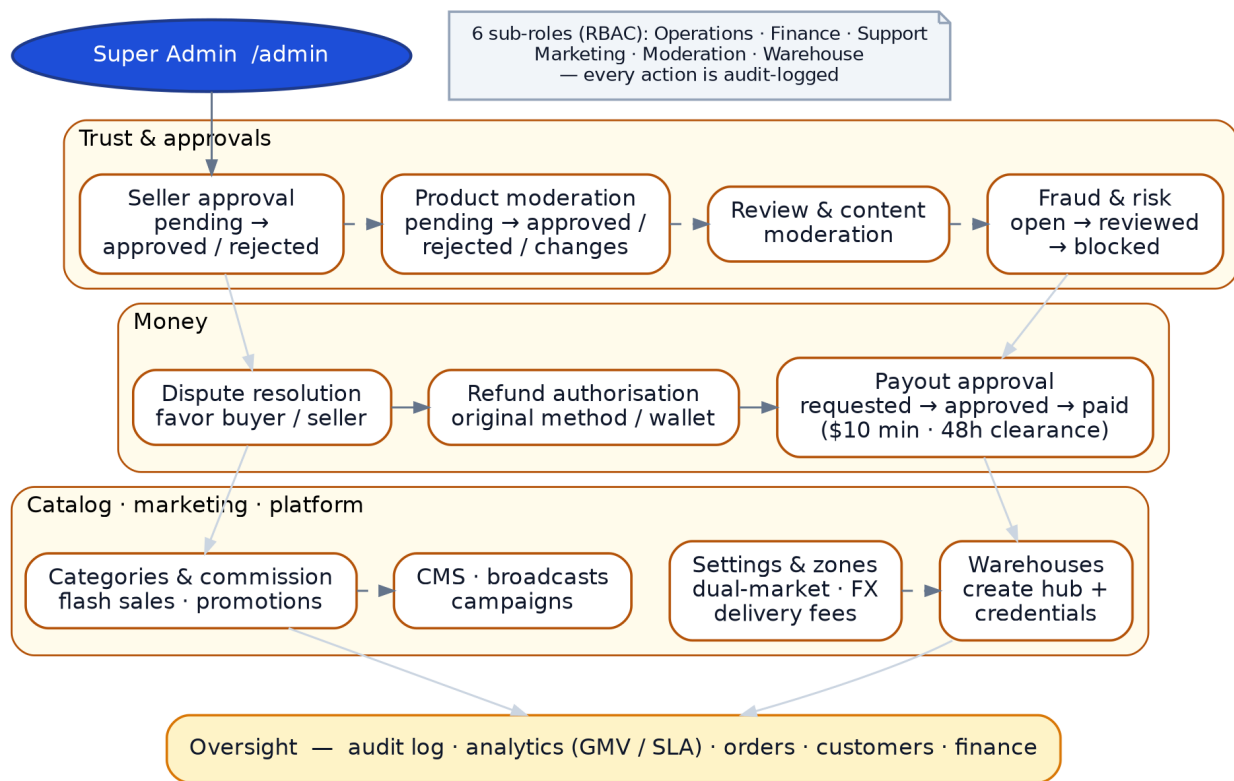


Figure 8 — Super Admin — governance grouped into trust & approvals, money, platform, and oversight.

Appendix · Status Reference

The lifecycle “states” used throughout the portals, collected for quick reference.

Order status

State	Meaning
pending → confirmed	Placed; payment confirmed
processing	Seller accepted / preparing
shipped → out_for_delivery	Dispatched; rider en route
delivered	Completed at the customer
cancelled / returned	Cancelled before delivery / returned after

Parcel & delivery status (warehouse + rider)

State	Meaning
inbound → received	Arrived at hub; scanned in
sorting → ready	Zoned; ready for a batch
dispatched	Handed to a rider in a batch
in_transit → out_for_delivery	Rider carrying / delivering
delivered / failed	Completed / attempt failed (re-dispatch)

Return, dispute & fraud status

Object	State progression
Return	requested → approved → in_transit → received → refunded (or rejected)
Dispute	open → seller_responded → resolved → closed
Fraud alert	open → reviewed → blocked (severity: low / medium / high)

Prepared from a full review of the application's portals, shared contracts and mock data. Flows describe intended production behaviour; the current build runs on mock data.